

VU KHANH CAO

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EDUCATION

Carleton University, Ottawa, ON

Sept 2023 – April 2028

Bachelor of Computer Science (Honours) – Artificial Intelligence and Machine Learning

GPA: 3.75 / 4.00

EXPERIENCE

Software Developer

May 2026 – Aug 2026

Bank of Canada

Ottawa, ON

- Pioneered the Bank of Canada's **first project** to modernize mission-critical applications **supporting over \$1 trillion in transactions annually** by designing **AI orchestration systems** for research, development, and multi-agent peer review.
- Designed agentic tooling for **code review and verification** within Azure CI/CD pipelines, improving compliance with **strict internal security and quality standards** while reducing manual validation effort.

Software Developer

May 2025 – Dec 2025

Ovenns Analytics Inc.

Ottawa, ON

- Designed and developed web platforms used to manage food preferences for restaurants, directly contributing across the application lifecycle from market research and product design to development and maintenance.
- Built **secure, scalable backend services** with Node.js, Express, and PostgreSQL to support multi-tenant access, reliable data handling, and maintainable API-driven workflows.

PROJECTS

Aether – [GitHub](#)

Feb 2026 - June 2026

- Led development of a Java-based automation tool reaching **2000+ users** and **\$10K MRR** within 1 month.
- Developed **resilient and tamper-proof authentication systems** applying dynamic class loading, bytecode obfuscation, and C native code transpilation while maintaining lightweight performance.
- Designed secure backend to handle software updates and licensing information with minimal latency using Supabase Edge Functions, **reducing server load by 90%**,

GURT – [GitHub](#)

Jan 2026

- Engineered a real-time robotic control system integrating embedded hardware, software, and secure WebSocket communication to support live video streaming and **sub-100ms** command execution.
- Developed a custom QNX 8 camera driver on Raspberry Pi 4 that increased video throughput by **10x**, improving reliability and responsiveness under real-time operating constraints.

JARVIS – [GitHub](#)

Jan 2025

- Built a low-cost physical-to-digital pipeline that converted 2D camera footage into web-ready 3D scenes using QNX, machine learning, and Gaussian Splatting, **reducing the barrier for entry** from **\$400 to \$20**.
- Containerized and deployed supporting services with Docker, **improving reproducibility, portability, and reliability** during development and demos.

SKILLS

Languages

Python, Java, JavaScript, TypeScript, C/C++, SQL

Libraries & Frameworks

React, Angular, Node.js, Express, Spring Boot, TensorFlow, OpenCV

Platforms & Technologies

Azure, AWS, Docker, Kubernetes, ArgoCD, OpenShift, REST APIs, Git